

DevOps Project Assignment – Containerized Payment Application

Objective

Build a **Python GUI** that consumes the existing Payment APIs and deploy the complete application using Docker.

Existing Setup

- **Payment Service** (REST APIs)
- **MongoDB**
- Running as **2 Docker containers**

Task

Develop a **Python/nodeJS/Any-based GUI** (python-Flask/Streamlit/Tkinter or similar) that:

- Consumes the existing Payment APIs.
- Performs payment operations.
- Displays API responses in a user-friendly interface.
- Handles errors gracefully.

Final Architecture

The application should run as **3 Docker containers**:

1. GUI
2. Payment Service
3. MongoDB

Deployment

Phase 1 – Local

- Dockerize the GUI.
- Use **Docker Compose** to run all three containers.
- Demonstrate end-to-end functionality.

Phase 2 – Cloud

Deploy the application on any cloud platform (AWS, Azure, GCP, Kubernetes, or equivalent).

Deliverables

- Python GUI source code
- Dockerfile
- Docker Compose configuration
- README with setup and deployment instructions
- Screenshots/demo of the working application

Testing:

GUI, Postman, FastApi